



Carnegie Mellon University
Master of
Software Engineering

Architecture views

LG Architecture Training Program

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Software architecture

*The software architecture of a system is the **set of structures** needed to reason about the system, which comprise **software elements, relations among them**, and properties of both.*

[Clements10]

Which structures?

- Which structures may an architecture consist of?
- Structures that describe
 - code units and their decomposition in submodules
 - code units and their inter-dependencies
 - components and how they interact at runtime
 - how artifacts are organized for deployment
 - the runtime infrastructure
 - data entities and their relationships in a domain model or data model
 - etc.

Architecture view

- A view is a representation of a structure, that is:

*A view is a representation of a set of system elements
and relationships among them.*
[Clements10]

- Not all system elements, some of them
- A view constrains the types of elements and the types of relations that can be represented in that view

*Views are intimately
related to viewpoints
or perspectives*



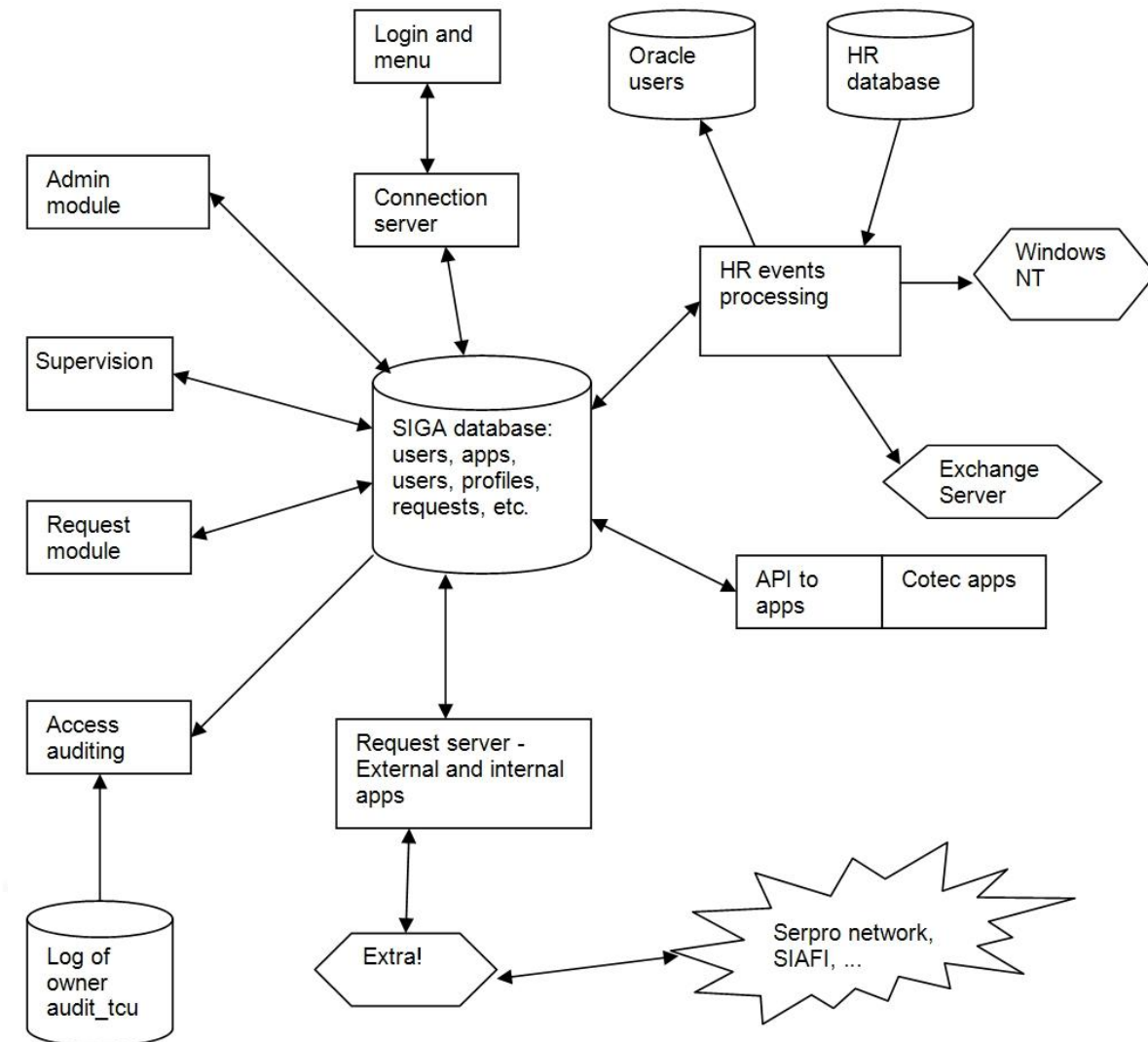
Eoin Woods

Boxes and lines in architecture diagrams

- Another way to ask what do the boxes and lines in a diagram mean would be:
 - What **types of elements** and **types of relations** does the diagram show?

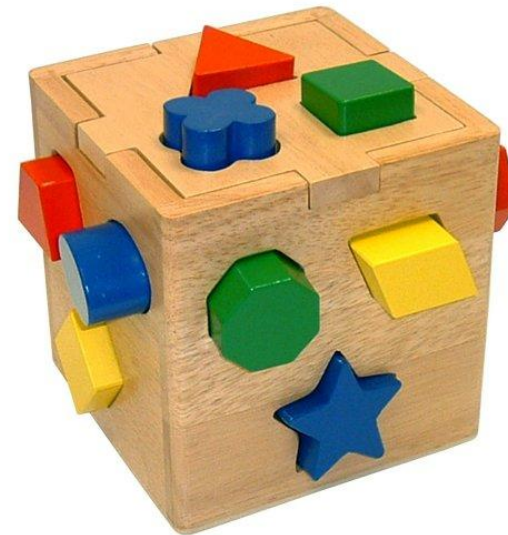
OR

 - What kind of view is this?



What views should I create in **my system**'s architecture documentation? 🤔

15:27 ✓



Views suggested by different authors



- Kruchten's 4+1
 - Logical view
 - Process view
 - Development view
 - Physical view
 - "Plus one" view
- Rozansky and Woods
 - Functional viewpoint
 - Information viewpoint
 - Concurrency viewpoint
 - Development viewpoint
 - Deployment viewpoint
 - Operational viewpoint
- RM-ODP
 - Enterprise viewpoint
 - Information viewpoint
 - Computational viewpoint
 - Engineering viewpoint
 - Technology viewpoint
- Philips CAFCR
 - Customer view
 - Application view
 - Functional view
 - Conceptual view
 - Realization view
- Arc42
 - Context and Scope
 - Building Block view
 - Runtime view
 - Deployment view
- TOGAF
 - Business architecture views
 - Data architecture views
 - Application architecture views
 - Technology architecture views
- Siemens Four Views
 - Conceptual view
 - Module interconnection view
 - Execution view
 - Code view
- Simon Brown's C4
 - Context diagram
 - Container diagram
 - Component diagram
 - Code diagram

*No fixed set
of views suits
all cases*



Paul Clements

What is the “right” set of views?

- *It depends!*
- Choose what views to create (and what views to spend more time on) based on:
 - The structures inherent to that software system
 - The stakeholders of the documentation and what would they use the view for



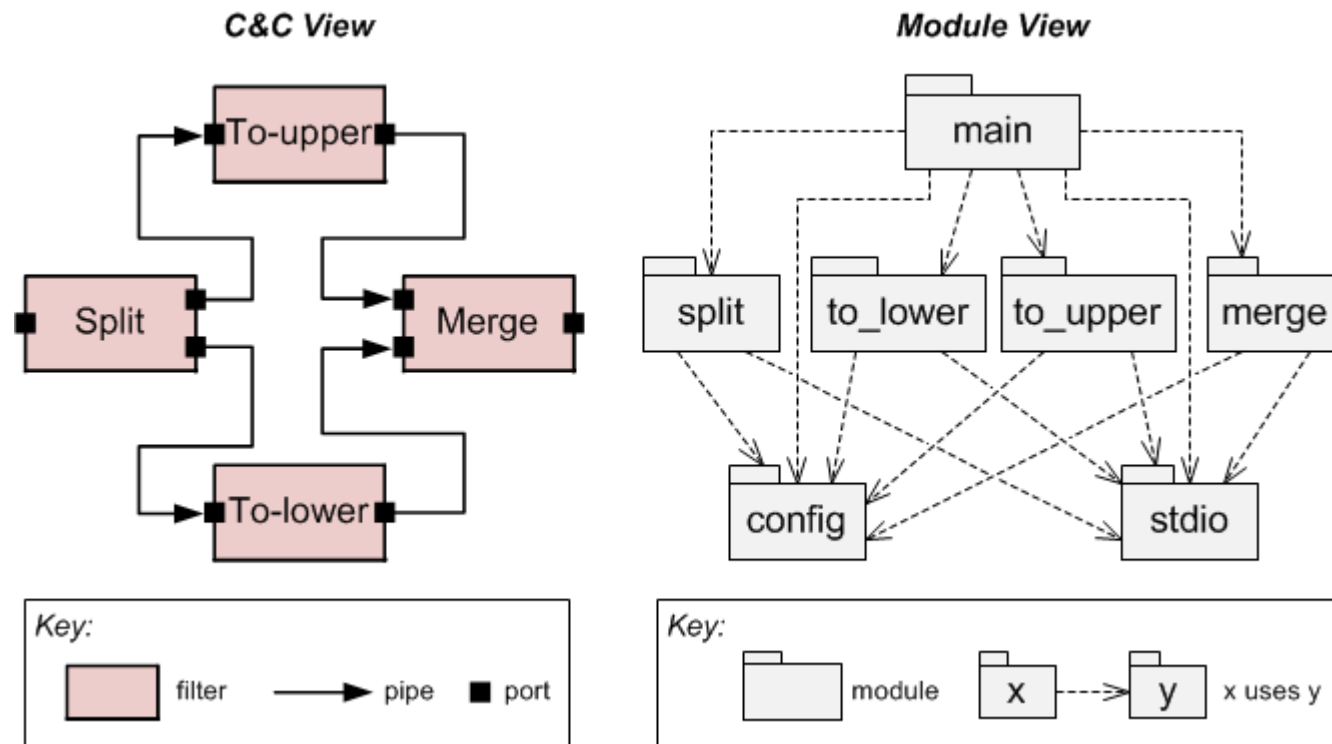
Categories of views

- We can minimally organize views in broad categories:
 - **Module views**: show the organization of implementation units (focus is code)
 - **Runtime views**: aka C&C views, show components and how they interact at runtime
 - **Other views**: include deployment structure, supporting hardware and runtime infrastructure

*In IClements101
the other views are
called Allocation
views*

Module vs component

- Module is an implementation unit
- Component is a software element with runtime presence



In general, the module to component mapping is many-to-many

Architecture style and architecture pattern

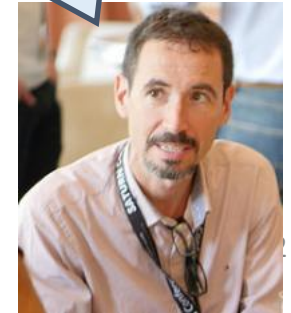
- We'll use the terms “style” and “pattern” indistinctively
- Examples of styles or patterns:
 - Layered
 - MVC
 - Microkernel
 - Publish-subscribe
 - REST
 - Microservice

Types of views and architecture styles

- The style/pattern used in the design can be used to name the architecture view describing that design
- Examples:
 - Layered view of the OSGi platform
 - Multi-tier view of the Adventure Builder system

Examples of software architecture documents:

- github.com/miyagis-forests/farmacy-food-kata
- wiki.sei.cmu.edu/sad





Questions?

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